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Large-scale Evaluation of Prokaryotic Annotation Tools Across Thousands of Species

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Abstract

Genome annotation is fundamental to prokaryotic genome sequencing, yet systematic evaluations guiding tool selection are lacking. We present the first large-scale investigation of four prominent open-source annotation tools (Prokka, Bakta, EggNOG-mapper, and PGAP) across 156,033 diverse genomes, including *Escherichia coli* strains for baseline performance, thousands of archaea and bacteria, frameshifted, and metagenome-assembled genomes. Bakta excelled in annotating high-quality bacterial genomes, while PGAP was better for archaeal and challenging genomes, including bacterial metagenome-assembled, fragmented, or contaminated samples. For biological functional Gene Ontology annotation, PGAP provided broader term coverage, whereas EggNOG-mapper offered more terms per feature. Our findings highlight tool-specific strengths crucial for selecting optimal solutions based on genome quality, taxonomy, and origin (e.g., MAGs). This study provides an evidence-based guide for users and informs future tool development.

Introduction

The rapid advancements in genomics and sequencing technologies have continuously improved our understanding of prokaryotic organisms, their genomic landscapes, and features. This progress is crucial as it underpins advancements across various disciplines, like biotechnology, biomedicine, and environmental science. As we move past well-studied genes and organisms, the demand for reliable and precise prokaryotic gene identification by annotation tools becomes increasingly important. Annotation is the cornerstone for gene identification, facilitating expression profiling and downstream applications by identifying and characterizing genomic features.

The complexity of the annotation process can be separated into two discernible phases. The initial annotation phase involves feature prediction leveraging machine learning algorithms like Hidden Markov model chains to differentiate coding from non-coding regions. Tools such as Prodigal[1], GeneMarkS-2[2], or Augustus are commonly employed for this purpose, while the latter is mainly used for eukaryotic organisms[3]. After initial annotation, filtering is applied to eliminate dubious or known false positives, although this can lead to the loss of important protein-coding genes below 100 amino acids (aa)[4]. Separately, distinct features, like RNA groups, are predicted using additional tools to improve accuracy.

In the subsequent annotation phase, the focus shifts to assigning a description for each coding sequence (CDS) through either homology or orthology interference [5,6]. Although orthology inference is considered more precise, it is computationally more demanding[7] . Various controlled vocabulary databases, including GO, KEGG, CARD, COG, and PFAM facilitate biological functional annotation of genes. Notably, each database has its specialty, such as CARD's focus on antimicrobial resistance genes[8] and PFAM's emphasis on protein domains and evolutionary relationships[9].

Although the annotation process is vital, there is a notable lack of comprehensive comparisons on tool performance and factors influencing it, such as source, taxonomy, or assembly quality. Existing tool comparisons narrow their focus on individual steps[10] or overlook recently developed pipelines[11].

Our study systematically evaluates popular, free, and open-source genome annotation tools to provide practical guidelines tailored to taxonomic context, desired annotation features, and error tolerance in Bacteria and Archaea (Fig.1). We focus on four widely used tools: Prokka[12], Bakta[13], EggNOG-mapper[14], and PGAP[6]. These tools represent a spectrum of annotation strategies, from Prokka's rapid, resource-efficient annotations to Bakta's extensive database coverage. EggNOG-mapper offers robust biological function and orthology-based annotation, including eukaryotes, but focuses on CDS features only. PGAP, used in-house by the National Center for Biotechnology Information (NCBI), stands out for its taxonomy-tailored pipelines, which ensure precise annotations but require prior taxonomic information.

To ensure broad applicability, we curated a diverse dataset using the Genome Taxonomy Database (GTDB), prioritizing its taxonomic accuracy over traditional NCBI datasets[15,16]. We made sure to incorporate scenarios reflecting real-world challenges, such as metagenome-assembled genomes (MAGs), simulated assembly issues (e.g., frameshifts), and temporal changes in database quality. These comprehensive assessments aim to guide tool selection based on specific needs and also highlight critical limitations to address in future tool developments.

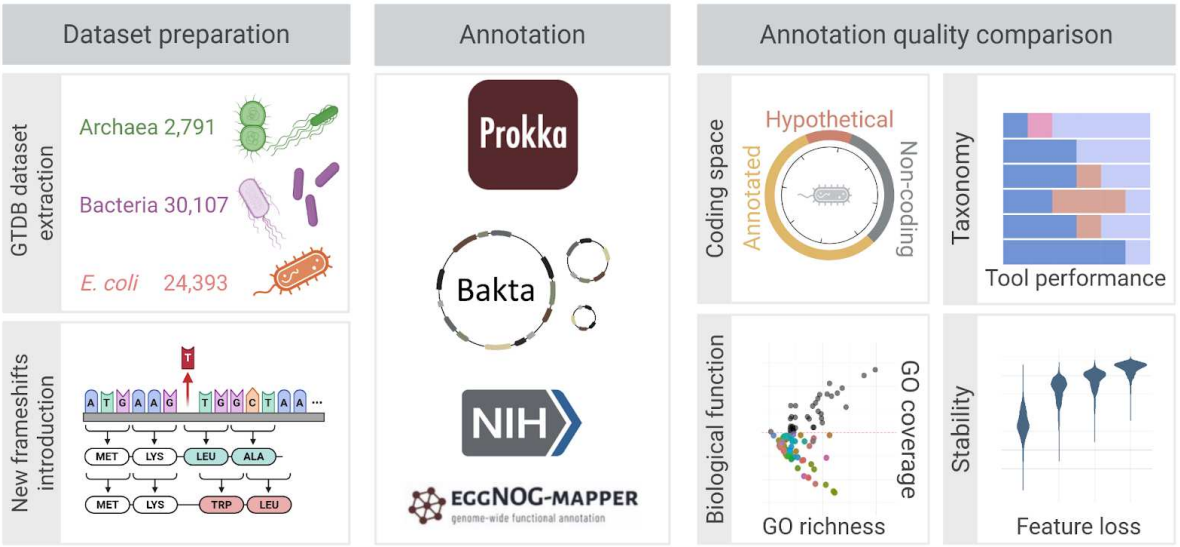

Fig.1: Large-scale Evaluation of Prokaryotic Annotation Tools Across Thousands of Species. The Genome Taxonomy Database (GTDB 207.0) was queried for representative Bacteria and Archaea species and all Escherichia coli strains. Additionally, genomes were generated from the representatives to include frameshifts of 0.5%, 1%, and 2%. All genomes were annotated using Prokka, Bakta, EggNOG-mapper, and PGAP with their recommended settings. The annotation quality between tools and genome conditions was compared based on multiple characteristics like coding space, detected feature length, Genome Ontology (GO) richness, and impact of frameshifts.

Results

Genome Data Acquisition and Annotation Tool Evaluation

For our large-scale benchmark of prokaryotic annotation tools, we annotated a total of 156,033 genomes to assess each tool's performance under various conditions. 29,761 bacterial and archaeal genomes were retrieved from the GTDB database (release 207.0) and downloaded from the NCBI database. We complemented this data by additionally downloading 24,393 different *Escherichia coli* strains. These *E. coli* strains were specifically included to establish a taxonomy-independent baseline performance for annotation quality, which allows comparisons to annotation influencing factors like taxonomy or frameshifts. Furthermore, we created 98,742 genomes to evaluate the impact of frameshifts on annotation quality by introducing pseudo-random genome-wide deletions (0.5%, 1%, and 2%) to a subset of the bacterial genomes, simulating sequencing errors. Additionally, 3,137 MAGs from different families were individually extracted and investigated (Methods: Datasets).

We selected Prokka, Bakta, EggNOG-mapper, and PGAP due to their free and open-source nature and overall popularity (Tab.1). For the analysis, we used each tool's recommended settings, except for the adjusted 4th translation table for the phylum *Mycoplasmatota* (Methods: Tools).

Tab 1. Overview of the benchmarked tool characteristics. CPU hours were used to adjust for different hardware settings during the benchmarking. The hardware settings used to minimize time and cost for this analysis are available in Methods: Tools; Workflow execution.

		Prokka	**Bakta**	**EggNOG-mapper**	**PGAP**
Scope	Coverage	Virus, Archaea, Bacteria, Mitochondria	Bacteria	Virus, Archaea, Bacteria, Eukaryotes	Archaea, Bacteria
	Annotated features (non-RNA)	CDS, Signal leader peptides	CDS, sORF, CRISPR, cis-regulatory regions, oriC/oriV/oriT, assembly gaps	CDS	CDS, CRISPR, Phage-related genes
	RNA features	tRNA, rRNA, ncRNA	tRNA, rRNA, ncRNA	N/A	tRNA, rRNA, ncRNA
	Main functional annotation databases	COG	COG, GO	COG, GO, KEGG pathways, Brite, EC	GO, EC
Prediction	Main prediction tools	Prodigal (CDS), RNAmmer (rRNA), Aragorn (tRNA), Infernal (ncRNA)	Prodigal (CDS), alignment-free sequence identification (< 29 aa proteins), tRNAscan-SE (tRNA), Aragorn (tmRNA), Infernal (rRNA, ncRNA)	Diamond/MMseqs2/ Prodigal (CDS)	ORFfinder/BLAST/ GeneMark-S2+ (CDS), Infernal (rRNA; ncRNA), tRNAscan-SE (tRNA)

	Prediction approach	Homology	Homology	Orthology	Homology
Database	Database size (unpacked)[Gb](+)	0.2	59.4	51.0	30.4
	Average tool update time (#)	No updates since November 2019	165 days	255 days	123 days
Performance	Median CPU time [min](*)	3.2	36.8	15.6	106.6
	Median RAM used [Gb](*)	0.6	8.6	41.7	4.5
Other	Requires taxonomy	Not required	Not required	Not required	Needs NCBI genus or species name
	Installation(-)	Conda, Docker, Singularity, Brew Integrated database that does not require additional handling	Conda, Docker, Singularity, PyPi Necessary independent database download and update	Conda, PyPi, script Necessary independent database download and tool setup	Docker, script The installation script downloads the database
	Execution	Single command, examples provided in the repository	Single command, examples provided in the repository	Complex single command, examples provided in the repository	Completion of the metadata YAML file is required before command execution
	Additional features		Genome visualization, JSON output		Auto-adjust taxonomy, Taxonomy checks, Robust annotation correction mechanism

+ Database size for the versions used for the analysis
# Checked from January 2001 to October 2024
* Across 9,842 different bacterial genomes; EggNOG-mapper was used with --dbmem flag for speed optimization
- Only installation methods officially recommended in the GitHub repository are listed.

We systematically assessed annotation quality across three key parameters: coding space, the count of RNA features (rRNA, tRNA, tmRNA), and biological function annotation using Gene Ontology (GO) terms (Method: Data analysis).

One of the primary metrics to assess each tool's ability to predict features was measuring the coding space. Coding space was separated into described and undescribed coding space, as opposed to the non-coding region, where no features were detected. Described coding space includes features with known descriptions, while undescribed coding space pertains to hypothetical proteins and features without any descriptions (which may consist of non-coding type features that a particular tool does not detect; Supplementary Fig.5). This approach minimizes the impact of feature size variations and naming discrepancies, unlike raw feature count. Ideally, tools should display a large, described coding space with minimal undescribed regions and consistent coding space sizes across the same well-studied species, e.g., *E. coli* (Method: Data analysis). The described features typically do not include information on biological processes and should not be confused with the biological function annotations of genes using GO terms.

We compared and quantified each tool's biological function annotation performance by using its assigned GO terms. Two metrics were measured: (1) GO coverage, the proportion of genes annotated with at least one GO term, and (2) GO richness, the number of GO terms per gene. Both metrics were considered, recognizing that, depending on the use case, a single detailed GO term can offer similar functional insights as multiple GO terms. Prokka was excluded from this analysis as it does not assign any GO terms.

Comprehensive Evaluation of Genome Annotation Tools Using *Escherichia coli* as a Model Reveals Tool-Specific Strengths

To assess baseline annotation performance, we evaluated all tools by using 24,393 different *E. coli* genomes, spanning 2,162 known serotypes (Methods: Datasets).

Coding space

Our initial analysis targeted the total coding space covering described features (yellow area, Fig.2a), undescribed features (red area, Fig.2a), and the non-coding space (gray area, Fig.2a). We also assessed feature lengths for inaccuracies, such as gene merging that leads to unusually long features[17] (Fig.2b). For reference, the average *E. coli* genome in our dataset spans 5,120,395 base pairs (bp) (standard deviation [SD] = 274,047 bp; Supplementary Fig.4).

When comparing total coding space (sum of described and undescribed space) across the 24,393 *E. coli* genomes, PGAP annotated the largest average total coding space, with a mean of 4,709,064 bp (SD = 301,104 bp), followed by Bakta (mean = 4,503,066 bp, SD = 229,128 bp) and EggNOG-mapper (mean = 4,498,985 bp, SD = 234,135 bp). Prokka exhibited the smallest total coding space (mean = 4,457,583 bp, SD = 223,485 bp).

When considering only the described coding space, Bakta had the largest average, followed by PGAP. Prokka's mean described coding space was substantially lower than that of the other tools (Bakta: mean = 4,485,441 bp, SD = 229,525 bp; PGAP: mean = 4,431,652 bp, SD = 210,208 bp; EggNOG-mapper: mean = 4,348,191 bp, SD = 193,514 bp; Prokka: mean = 3,614,941 bp, SD = 102,652 bp). However, it exhibited the lowest variability, possibly due to its limited feature annotation, as reflected in the smaller described coding space.

When comparing the preferred small undescribed coding space, Bakta demonstrated the smallest across all genomes with the lowest mean and deviation (mean = 11,962 bp, SD = 17,084 bp), followed by PGAP, EggNOG-mapper, and Prokka (PGAP: mean = 109,812 bp, SD = 30,664 bp; EggNOG-mapper: mean = 150,794 bp, SD = 66,604 bp; Prokka: mean = 841,628 bp, SD = 149,351 bp). On a side note, Prokka and EggNOG-mapper generated longer mean undescribed features (Fig.2b red). EggNOG-mapper may have assigned long noncoding features like assembly gaps as hypothetical proteins (Supplementary Fig.5), which resulted in a bimodal undescribed coding length distribution (Fig.2b, red EggNOG-mapper). For all comparative analysis between the tools, see Supplementary Tab. 1-7.

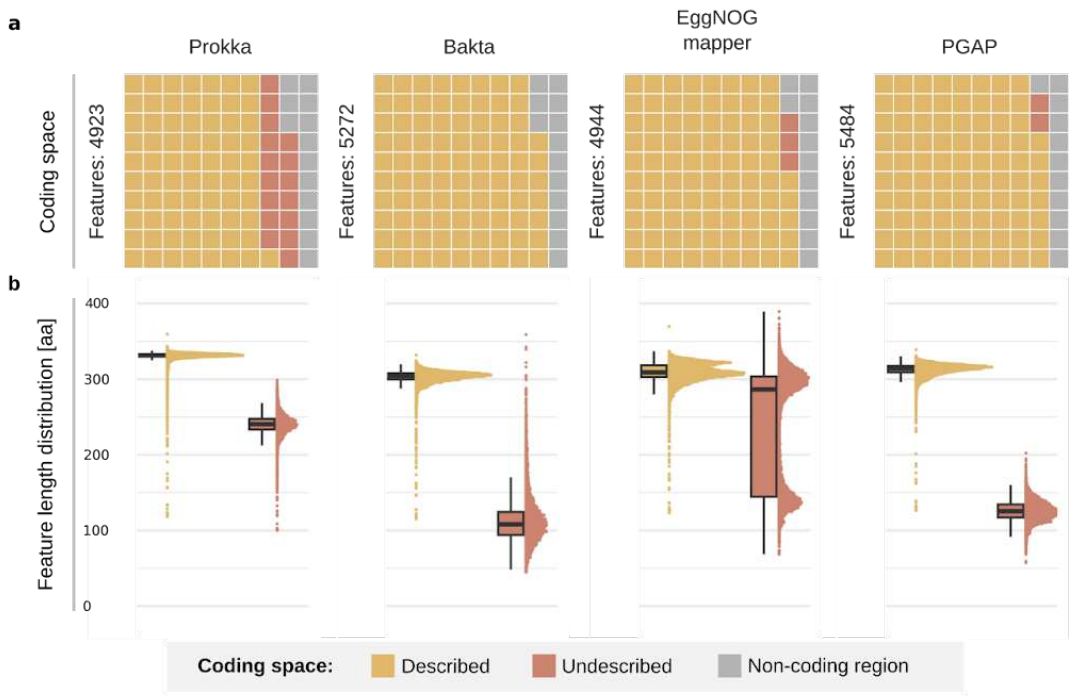

Fig.2: E. coli coding space and gene length distribution. (a) Coding space: The average coding space detected by the tools is categorized into described coding space (yellow), undescribed coding space (red), and non-coding regions (gray). Each square represents 1% of the total genome. The mean feature count is noted on the left. (b) Coding length distribution: The average amino acid length distribution of features for described (yellow) and undescribed (red) coding space is shown.

RNA features

Next, we evaluated the annotation of RNA features. *E. coli* contains seven operons, each composed of 16S rRNA, 23S rRNA, and 5S rRNA genes, with one additional 5S rRNA, totalling 22 rRNA genes[18,19]. For simplicity, we considered only the total detected rRNA count, without distinguishing between complete or incomplete operons. EggNOG-mapper was excluded from this analysis as it does not annotate rRNA features.

None of the tools consistently annotated all rRNA operons across the *E. coli* strains. On average, PGAP identified the highest count of rRNA genes (mean = 13.89, SD =

8.36), followed by Prokka (mean = 9.76, SD = 6.37) and Bakta (mean = 9.60, SD = 6.27). The requirement for at least one complete operon was not fulfilled in 781 cases with Prokka, 779 with Bakta, and 451 with PGAP. In genomes with fewer than 10 contigs, all tools consistently annotated a similar high mean of rRNA (Prokka: mean = 21.97, SD = 1.33; Bakta: mean = 21.96, SD = 1.27; PGAP: mean: 21.95, SD = 1.20; Supplementary Fig.6a). However, in samples with more fragmented genomes (> 10 contigs), the total annotated rRNA count decreased substantially. PGAP maintained the highest count in these cases (mean = 12.96, SD = 8.33), compared to Prokka and Bakta (Prokka: mean = 8.34, SD = 5.08; Bakta: mean = 8.17, SD = 4.89).

E. coli contains 79 tRNA genes[20], and PGAP detected a similar count of these features (mean = 78.74, SD = 12.47). Prokka and Bakta detected, on average, three more features (Prokka: mean = 81.15, SD = 12.53; Bakta: mean = 81.72, SD = 12.59) (Supplementary Fig.6b).

Bacteria are generally expected to have a single tmRNA gene per genome[35]. All evaluated tools demonstrated a mean tmRNA value of 1 (SD: Bakta: 0.08; PGAP: 0.09; Prokka: 0.10). However, Prokka and Bakta failed to identify at least one tmRNA in 62 cases, while PGAP missed it in 15 genomes. Furthermore, Prokka identified more than one tmRNA in 121 instances, Bakta in 83, and PGAP in 127.

Biological functional annotation

To evaluate the performance of functional annotation, we analyzed the GO coverage and GO richness. EggNOG-mapper exhibited the highest GO coverage, with a mean of 66.07% (SD = 4.59%), comparable to Bakta (mean = 63.88%, SD = 5.25%). PGAP showed the lowest average GO coverage of 36.12% (SD = 1.95%). For GO richness, EggNOG-mapper led with a mean of 37.00 terms per gene (SD = 0.48), compared to Bakta (mean = 4.00, SD = 0.02) and PGAP (mean = 2.00, SD = 0.00). It is important to note that EggNOG-mapper identified fewer genes than PGAP (Fig.2a), which might influence both GO richness and coverage performance (Supplementary raw data).

Investigating Annotation Performance Across 29,761 Bacterial and Archaeal Species: Taxonomy-Related Disparities Among Tools

Using the core performance metrics established for *E. coli*, we evaluated the impact of taxonomic variance on annotation quality across Bacteria and Archaea. This section covers 2,791 archaeal (20 phyla) and 26,970 bacterial (163 phyla) representative species. We excluded 3,137 bacterial MAGs and analysed them separately in the “Assessing Annotation Stability” section, because of their later outlined properties affecting annotation performance. However, due to the low amount of Archaeal non-MAGs (760 non-MAGs to 2,031 MAGs), we decided to include these MAGs to maintain a meaningful Archaea representation (Methods: Datasets).

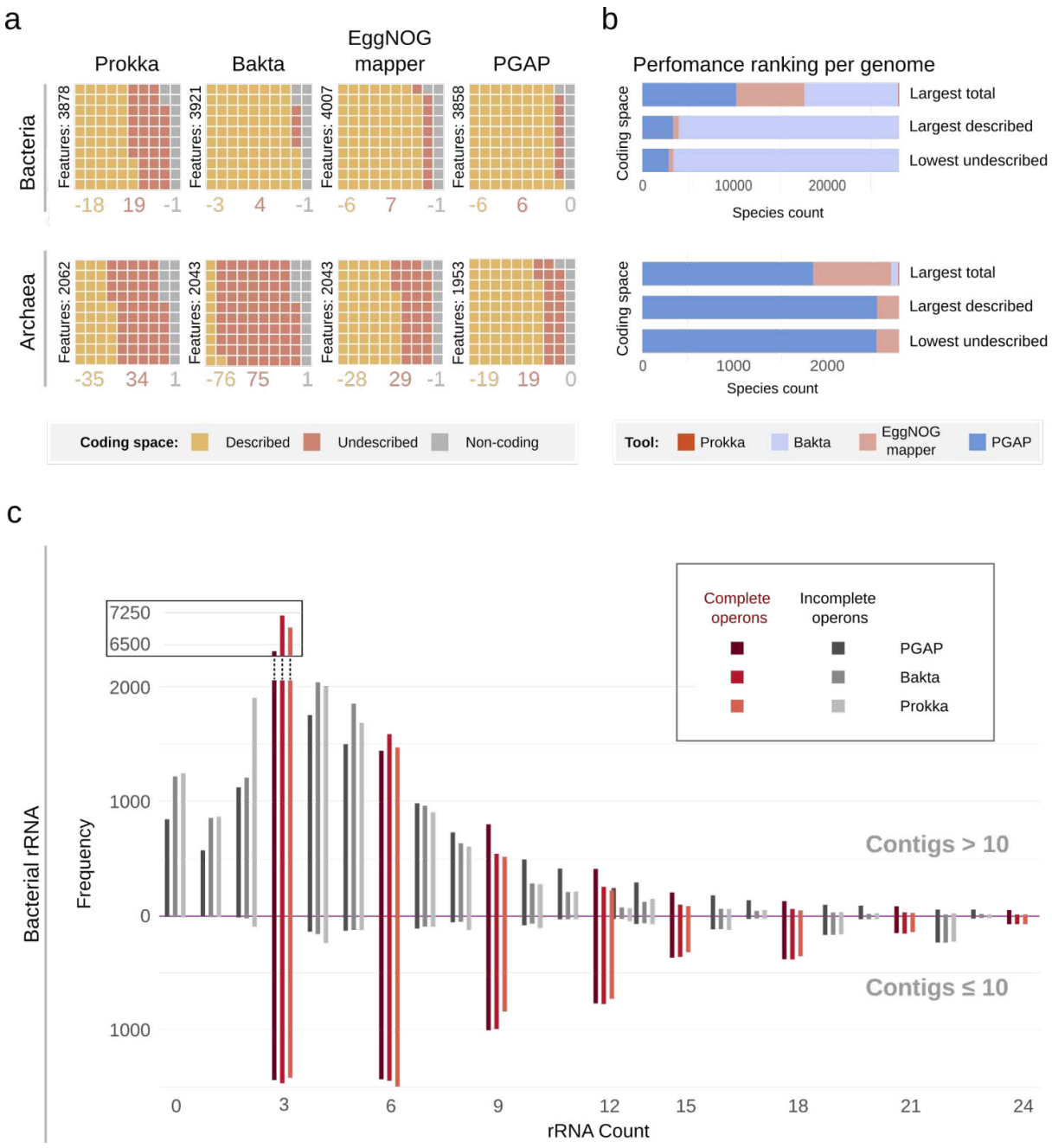

Fig.3: Taxonomy performance analysis. (a) Coding space: Breakdown of the average coding space detected by each tool, categorized into described coding space (yellow), undescribed coding space (red), and non-coding regions (gray), separated into Bacteria and Archaea. Each square represents 1% of the total genome. The mean feature count is noted on the left. The annotated gain or loss to the E. coli baseline is outlined in colored numbers below each plot (e.g., a yellow -18 means that this tool annotated 18% less described coding space compared to the baseline). (b) Performance ranking per individual genome: The bar charts highlight the number

of times a tool has performed best in one of the three categories for each species (largest total, largest described, and lowest undescribed coding space). The difference in performance between the best and second-best tools was not considered by this approach. (c) Distribution of rRNA counts in complete and incomplete operons across each tool in Bacteria. The upper panel displays the more fragmented genomes, characterized by having more than 10 contigs. In contrast, the bottom panel shows the less fragmented samples, with fewer than 10 contigs. Complete operons, containing multiples of 3 rRNA genes, are highlighted in red, while incomplete operons are displayed in gray. Only rRNA count up to 24 is shown for plot clarity. The Archaea counterpart is presented in Supplementary Fig.14.

Coding space

Coding space was analyzed within Archaea and Bacteria using the same metrics as the baseline: total, described, and undescribed coding space. The average total coding space across all tools ranged from 88.01% to 88.85% of genome length, aligning with the 87% reported in the literature[21]. While the proportion of total coding space relative to genome size remained consistent with baseline performance, we observed a notable shift from described to undescribed coding space, particularly in Archaea (Fig.3a). To provide “taxonomic-based” annotation recommendations, we identified the best-performing tool for each genome based on total and described coding space metrics (Fig.3b). Detailed species-level results covering each tool's performance are available in Supplementary raw data and Supplementary Fig.10 and 15.

Total Coding Space: Among 26,970 bacterial species, PGAP annotated the largest total coding space for 9,872 species, followed by Bakta (9,860 species), EggNOG-mapper (7,162 species), and Prokka (76 species) (Fig.3b, Supplementary Fig.9-10a). In 2,791 archaeal genomes, PGAP showed the highest outcome in 1,859, EggNOG-mapper in 847, Bakta in 77, and Prokka in 8 cases (Fig.3b, Supplementary Fig.15a). These results suggest that no single tool consistently maximized total coding space across all taxa. While some genus-level performance trends were noted, no meaningful taxonomic recommendations could be made.
Described Coding Space: Next, we checked how much of the total coding space is described. For bacteria, Bakta had the largest described coding space in 23,122

species, far outpacing PGAP (3,252 species) and EggNOG-mapper (596 species), with Prokka not leading for any species (Fig.3b, Supplementary Fig.10b). These results position Bakta as a strong general recommendation for annotating the described coding space in bacteria. In archaea, PGAP led in 2,550 species and EggNOG-mapper in the remaining 241 species, while neither Bakta nor Prokka led in any cases (Fig.3b, Archaea).

RNA features

The RNA section largely mirrors the structure of the baseline performance analysis. The results were similar in Bacteria, while in Archaea, these counts were lower across all tools (Supplementary raw data, Supplementary Fig.14). To assess the correctness of the rRNA annotation, we investigated if they appear as full rRNA operon (multiples of 3 rRNA; Fig.3c, red bars).

Among non-fragmented genomes (Contig ≤ 10), all tools exhibited high proportions of rRNA counts matching the complete operons (Bacteria: Bakta: 79.0%, PGAP: 78.5%, Prokka 75.2%; Archaea: Bakta: 67.2%, PGAP: 67.4%, Prokka 55.4%), which declined in highly fragmented genomes (Contig > 10; Bacteria: Bakta: 50.4%, PGAP: 49.3%, Prokka: 47.9%; Archaea: Bakta: 27.9%, PGAP: 35.4%, Prokka: 25.2%). For highly fragmented genomes, PGAP often predicted the highest number of rRNA sequences, suggesting its ability to identify all parts of a fragmented rRNA gene split across multiple contigs (Supplementary Fig.11, Tab.8-10). In comparison to tRNA in the taxonomical variable group, Prokka and Bakta predicted a higher number of tRNAs (Prokka: mean = 53.20, SD = 18.30; Bakta: mean = 53.20, SD = 18.0) compared to PGAP (PGAP: mean = 51.6, SD = 17.5). Moreover, considering the minimum number of tRNAs required to cover 20 essential amino acids as proposed by Land *et al*. (2015), Bakta failed to meet this requirement in fewer cases (352) than Prokka and PGAP (Prokka: 413, PGAP: 441). In Archaea, Bakta and PGAP produced consistent results (Bakta: mean = 38.54, SD = 10.58; PGAP: mean = 38.34, SD = 9.68; Prokka: mean = 37.47, SD = 10.58).

While bacterial tmRNA results were consistent with the baseline, we observed tmRNA sequences in 51 archaeal samples, where it is not known to occur. Following Nawrocki *et al.* (2025), we attribute this to metagenome contamination. However, in

10 Bakta and 4 Prokka cases, it was found in non-metagenome samples (Supplementary raw data).

Biological functional annotation

Building upon the preestablished biological function annotation baseline in *E. coli*, we observed a significant decline in Gene Ontology (GO) term coverage when extending the analysis beyond *E. coli* to Bacteria and Archaea. This decline was evident for annotations generated by Bakta and EggNOG-mapper.

In Bacteria, PGAP consistently provided the highest GO coverage (mean = 31.20%, SD = 6.92%), outperforming both EggNOG-mapper (mean = 11.40%, SD = 11.6%) and Bakta (mean = 2.40%, SD = 4.62%). Notably, Bakta did not exceed PGAP's GO coverage in any bacterial family. However, EggNOG-mapper performed better than PGAP in ten taxonomic families, including clinically relevant groups such as *Enterobacteriaceae*, *Mycobacteriaceae*, and *Bacillaceae* (Fig.4). In Archaea, the decline in GO coverage was even more pronounced. PGAP maintained the highest performance (mean = 20.03%, SD = 3.30%), while EggNOG-mapper (mean = 5.80%, SD = 4.44%) and Bakta (mean = 0.13%, SD = 0.11%) performed considerably worse. Across all tools, incorporating taxonomic variation reduced the average GO coverage, with PGAP remaining the most reliable option overall.

GO richness remained comparable to the baseline established in *E. coli,* in contrast to GO coverage. EggNOG-mapper demonstrated the highest mean GO richness in both Bacteria (mean = 35.40, SD = 10.60) and Archaea (mean = 38.70, SD = 14.30), far surpassing Bakta (Bacteria: mean = 3.56, SD = 1.18; Archaea: mean = 2.00, SD = 0.03) and PGAP (Bacteria: mean = 2.03, SD = 0.16; Archaea: mean = 2.00, SD = 0). This result reinforces EggNOG-mapper’s utility in generating richer functional annotations, albeit with the previously mentioned lower overall GO coverage compared to PGAP.

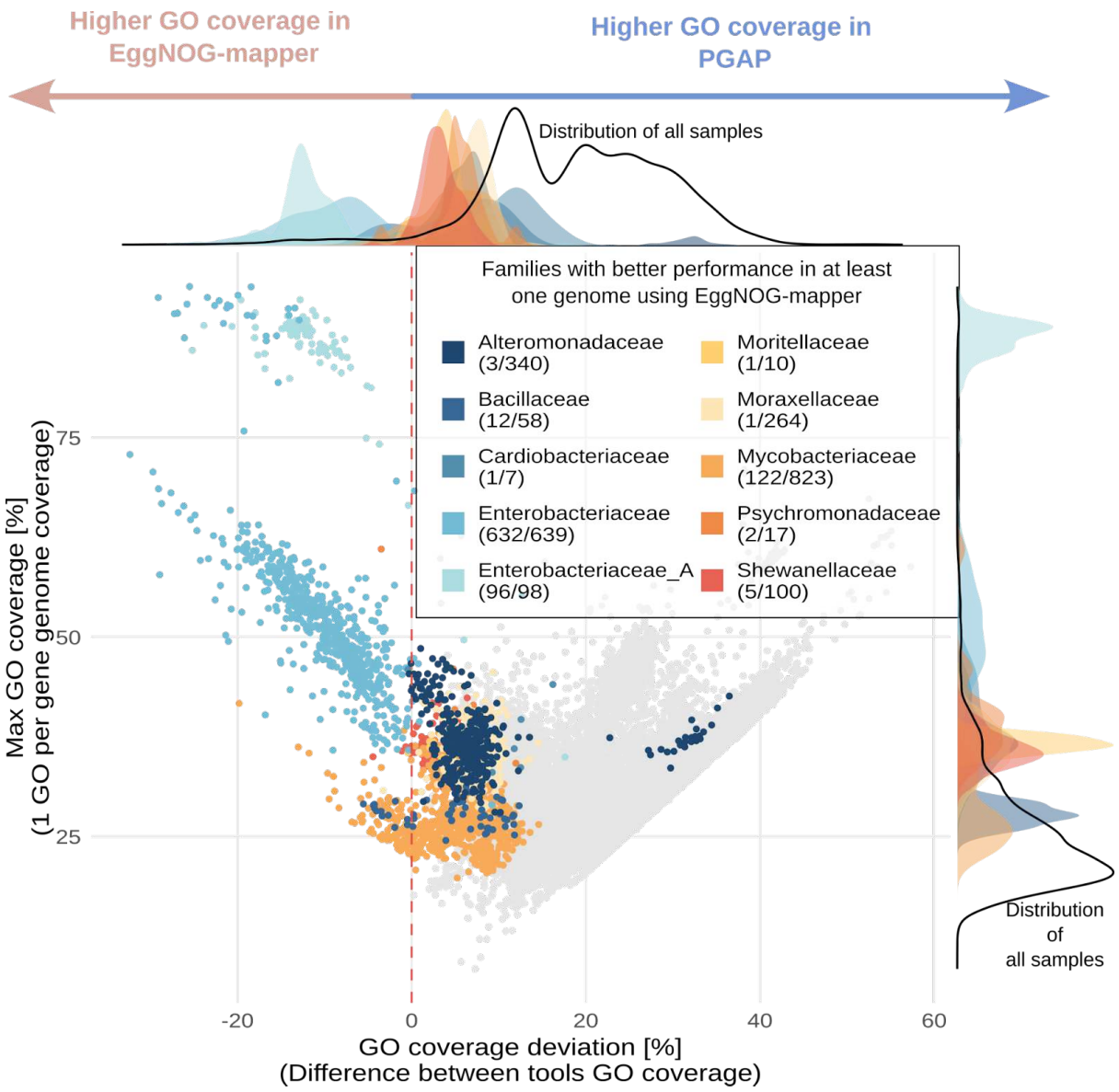

Fig.4: *Comparison of Gene Ontology (GO) term coverage between PGAP and EggNOG-mapper across 26,970 bacterial genomes. Each point represents a genome, with the y-axis displaying the maximal GO coverage across both tools for that genome, and the x-axis showing the deviation in GO coverage between EggNOG-mapper and PGAP. The red vertical line indicates equal GO coverage between the two tools. The black outline line on the marginal plot shows the distribution of the parameters across all samples. Points to the right of the red line (positive deviation) represent genomes where PGAP had higher GO coverage than the average. In comparison, points to the left (negative deviation) represent genomes where EggNOG-mapper had higher GO coverage. For example, a point at +20% on the x-axis indicates PGAP's GO coverage was 20% above the average,*

while -10% indicates that EggNOG-mapper's GO coverage was 10% higher. Gray dots correspond to genomes from bacterial families where PGAP consistently outperformed EggNOG-mapper across all members. Colored dots indicate genomes from families where EggNOG-mapper outperformed PGAP in at least some cases. The number of such cases and the total number of genomes in each family are shown in the legend.

Assessing Annotation Stability: Metagenome-assembled Genomes, Frameshift Sensitivity and the Role of Temporal Delays in Genome Annotation Performance

Metagenome-assembled Genomes

Identifying novel microbes often relies on environmental samples and non- or hard-to-cultivable organisms, making isolates difficult to obtain. MAGs are, therefore, crucial; however, their correct assembly is complicated by factors such as strain mixtures, chimeric assemblies, incomplete sequences, truncated genes, and frameshifts[23]. We evaluated MAG annotation performance using 3,137 high-quality bacterial MAGs, all from different families (as defined by GTDB) to cover a broad taxonomic range. The selected MAGs exhibited lower completeness (mean = 85.22%, SD = 11.94%) and higher contamination levels (mean = 1.67%, SD = 1.59%) compared to non-MAGs (Supplementary Fig.16). Archaeal MAGs were not investigated because they were included in the previous section due to the low number of non-MAGs.

The comparison between MAGs and the previously established *E. coli* baseline revealed a substantial, yet non-uniform, shift from described to undescribed coding space, within each tool (Fig.5a, b). This shift was more pronounced than the previously described taxonomic impact. Bakta experienced the most significant increase in undescribed coding space, whereas PGAP now provides the largest described and lowest undescribed coding space on average for MAGs. A substantial decrease in the described coding space was also observed in EggNOG-mapper, but to a lesser extent than Bakta. In contrast, Prokka displayed consistent behavior

across samples and exhibited the smallest observed differences from the baseline. Additional gene length analysis is presented in Supplementary Fig.17-18.

The rRNA measurements in MAGs generally followed the trends seen in the baseline measurements. PGAP identified more rRNA genes compared to other tools (PGAP: mean = 2.10, SD = 1.96; Prokka: mean = 1.49, SD = 1.52; Bakta: mean = 1.48, SD = 1.46). Regarding the biological functional annotation, all tools GO richness remained stable when compared to the baseline. The GO coverage of PGAP decreased to a similar level as the taxonomic variable group (mean = 31.47%, SD = 5.4%). In contrast, the decrease of other tools was higher (EggNOG-mapper: mean = 6.78%, SD = 5.28%; Bakta: mean = 0.21%, SD = 0.92%).

Frameshift sensitivity

During the processing of genome data, NCBI classifies samples as problematic if more than 30% of the protein-coding genes exhibit frameshifts (https://www.ncbi.nlm.nih.gov/datasets/docs/v2/policies-annotation/genome-processing/genome_notes/). We investigated the impact of frameshifts on annotation performance for each annotation tool by creating 98,742 frameshifted genomes. This subset was created by randomly deleting 0.5%, 1%, and 2% of nucleotides using 32,914 different genomes (Methods: Frameshifts). There are two aspects to consider for any given feature with a frameshift: first, the tool may fail to detect it or not fully annotate it, increasing the non-coding space; second, it may detect the sequence but split it into multiple features (increasing protein-coding gene count).

The total coding space decreases progressively as the deletion rate increases in all tools, but to a noticeably lesser extent in PGAP, compared to the original sequence (Fig.5c).

First, the large decrease in total coding space for Prokka, Bakta, and EggNOG-mapper indicates that frameshifts are causing parts of features to be missed. Yet, there was a twofold increase in the total number of features from the original sequence to the 0.5% noise level, with feature count stability between 0.5% and 1%, followed by a decrease in feature count at 2% for all three tools (Fig.5d). This suggests that these tools can detect certain split fragments of features but not all, leading to a gradual reduction in total coding space as noise increases while

feature counts go up. However, at a 2% deletion rate, the feature count drops again, indicating that features become too broken up to be detected.

In contrast to the other three tools, PGAP exhibits a distinct pattern. The overall total coding space decreases less dramatically, and the feature count also increases to a lesser extent compared to the other tools. This combination of data suggests that PGAP is more effective at identifying most of the frameshifted parts while simultaneously annotating them into a single feature, thereby avoiding multiple feature descriptions for the same original feature (Supplementary Fig.19). Additionally, the 2% noise rate did not result in a decrease in feature count, as previously observed in the other tools. Thus, PGAP is still able to annotate the fragmented features. Specifically, PGAP showed smaller feature increases at the 0.5% level, followed by slight increases at 1%, and stability at 2%, with low variability between samples (Fig.5d). This demonstrates the benefit of using ORFfinder and ProSplign frameshift detection included in PGAP[6,24], compared to the use of only Prodigal by Prokka and Bakta.

We found no significant shifts in rRNA and tRNA counts across different noise levels, likely due to the relatively small proportion of occupied coding space of these features and the requirement for splits to occur within these particular sequences.

Temporal impact

Annotation tools rely on databases like InterPro for gene function descriptions, which are regularly updated with new proteins (Supplementary Fig.20). These database changes require the user to update their annotation tool to refresh their database (Tab.1). We were able to evaluate this temporal impact on annotation quality, by comparing our current PGAP annotation feature count with older annotation information stored in the GTDB metadata (Supplementary Fig.22a,b,c). On average, our more recent PGAP annotations detected 139.8 (SD = 134.79) more features than the previous annotation, highlighting the importance of regular updates. As the exact time of the older annotation was not available, further analysis could not be performed based on the available data. There was no change in rRNA counts between the re-annotations.

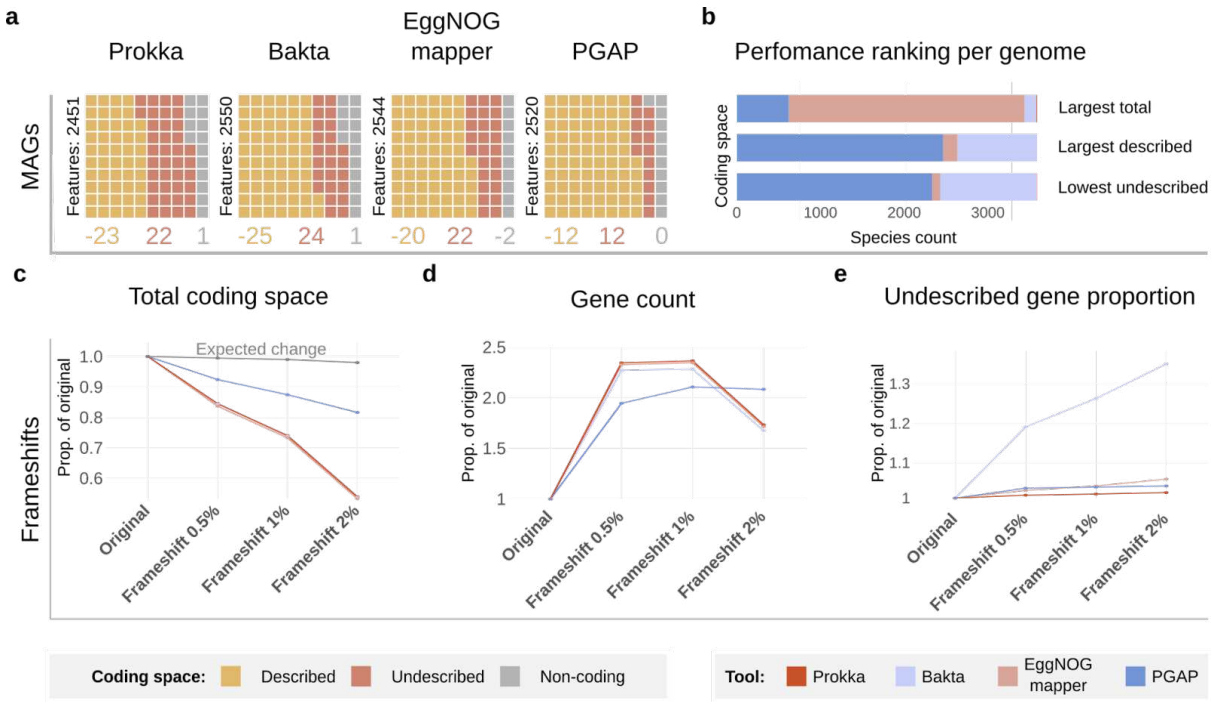

Fig.5: *Impact of bacterial metagenome-assembled genomes (MAG)s and frameshifts on annotation quality. MAGs: (a) Breakdown of the average coding space detected by each tool, categorized into described coding space (yellow), undescribed coding space (red), and non-coding regions (gray). Each square represents 1% of the total genome. The mean feature count is noted on the left. The annotated gain or loss to the E. coli baseline is outlined in colored numbers below each plot (e.g., a yellow -23 means that this tool annotated 23% less described coding space compared to the baseline). (b) Performance ranking per individual genome: The bar charts highlight the number of times a tool has achieved the best performance in one of the three categories for each species (largest total, largest described, and lowest undescribed coding space). This approach did not account for the performance difference between the best and second-best tools. Frameshifts: The effect of frameshifts on gene annotations was evaluated under three genome-wide deletion conditions (0.5%, 1%, and 2%) in Bacteria. We displayed mean proportion change to the original sequence of (c) retained coding space, (d) gene count, and (e) proportion of hypothetical features compared to the mean original value across four annotation tools (Prokka, Bakta, EggNOG-mapper, and*

PGAP). The total coding space plot (c) also includes expected differences in coding space based on the different deletion proportions.

Discussion

The rapid growth in sequenced prokaryotic species and the discovery of new proteins underscore the importance of accurate and comprehensive genome annotation to enable meaningful biological downstream analysis. Given the diversity of prokaryotic microorganisms and the varying quality of genomic data, understanding the strengths and limitations of different annotation tools is paramount. Thus, we explored the capabilities and differences of various gene annotation tools designed for prokaryotic annotation, namely Prokka (15,000 citations[12]), Bakta (800 citations[13]), EggNOG-mapper (3,100 citations[14]), and PGAP (5,800 citations[6]), across a large and diverse dataset.

Even after analysing and annotating over 156,033 genomes (including *E. coli* strains, bacteria, archaea, and simulated genomes), we did not observe a clear "winner" among the evaluated tools. Instead, our findings reveal specialized applications depending on genome properties and research objectives. This supports a "choose the right tool for the right genome and goal" approach, rather than a “one-size-fits-all” strategy. We thus outline recommendations based on genome characteristics and annotation priorities below (Visual decision tree in Supplementary Fig.24).

For typical bacterial isolates, our analysis revealed a nuanced tool recommendation between Bakta and PGAP. While taxonomic variation did not lead to clear genus-level recommendations for maximizing total coding space, Bakta consistently annotated the largest described coding space across many bacterial species. This suggests Bakta is highly effective at assigning known functional descriptions to coding sequences in bacterial isolates. PGAP, on the other hand, annotated the largest total coding space in a similar number of bacterial species and demonstrated reliable rRNA, tRNA, and tmRNA detection in fragmented bacterial genomes. Therefore, Bakta is strongly recommended for bacterial isolates when maximizing the described coding space, which is the primary goal. In contrast, PGAP is

recommended for maximizing total coding space in fragmented genomes e.g. from short-read assembly or ensuring robust detection of RNA features.

In Archaea, our results indicate a more apparent preference for PGAP. Across 2,791 archaeal genomes (including metagenome-assembled genomes (MAGs)), PGAP consistently annotated the largest total and described coding space, significantly outpacing other tools. PGAP also performed better in detecting rRNA in more fragmented archaeal genomes. This suggests PGAP is the most robust option for archaeal annotation, regardless of whether the genome is an isolate or a MAG.

The annotation of "problematic genomes," such as MAGs or those with significant fragmentation, errors, or potential contamination, presents unique challenges. PGAP showed the best stability in MAG annotation, by providing the largest described coding space and the lowest undescribed coding space on average. Furthermore, our targeted frameshift analysis, simulating problematic sequencing errors, revealed that PGAP is significantly more resilient to sequence errors, maintaining a more consistent annotated coding space and feature count compared to Bakta, EggNOG-mapper, and Prokka, which showed greater disruptions. Considering these factors, PGAP is a strongly recommended tool for annotating potentially problematic genomes, where issues such as fragmentation, contamination, and sequencing errors are more likely to occur.

Biological function annotation quality, specifically Gene Ontology (GO) term assignment, varied, but PGAP and EggNOG-mapper stand out as clear recommendations. PGAP consistently provided the highest GO coverage (the proportion of genes with any GO term assigned) across Bacteria and Archaea, including MAGs, albeit with lower GO richness (the number of unique terms per annotated gene) than EggNOG-mapper. However, EggNOG-mapper showed better GO coverage than PGAP in specific bacterial families, including some with clinical relevance, which should be considered when choosing. Therefore, the choice for biological function annotation depends on the particular goal. If maximizing the number of genes with any function assignment is critical (coverage), PGAP is generally recommended, especially for diverse or lower-quality genomes. If obtaining a wider variety of GO terms for annotated genes is prioritized (richness),

EggNOG-mapper performs better, particularly in less problematic genomes and specific taxa. Users might consider running both tools.

In conclusion, our comprehensive evaluation of prokaryotic genome annotation tools highlights that no single tool is universally optimal. The ideal choice is contingent upon several factors, including the taxonomic affiliation of the organism, the quality and fragmentation of the genome assembly (e.g., isolates vs. MAGs, presence of errors), and the specific research goal (e.g., maximizing total coding space, maximizing described function, reliable RNA annotation). Our research emphasizes the importance of considering taxonomy, sequencing technologies, sample quality, and database updates when using, benchmarking, or developing annotation tools. Understanding these characteristics is essential for both tool users, who must select the most appropriate tool for their specific data and question, and developers, who can use these insights to improve future algorithms and database integration. Therefore, this work offers valuable insights for both groups by demonstrating overall patterns in tool behavior and potential problem areas, which will continue to be relevant in the future for prokaryotic genomics.

Methods

Datasets

Data querying and extraction

All analyzed prokaryotic genomes were obtained by querying the release 207.0 of Genome Taxonomy Database (GTDB) (https://gtdb.ecogenomic.org), contained in the release files bac120_metadata_r207.tsv for Bacteria and ar53_metadata_r207.tsv for Archaea records. GTDB files were also used for sample metadata. Genome assemblies were then downloaded from NCBI Datasets repository via NCBI Datasets command line tool v.14.23.0 using the command "`datasets download genome accession –inputfile genome_list.txt --dehydrated`", whereas "genome_list.txt" contains GTDB database keys "ncbi_genbank_assembly_accession" of selected samples. Next, the samples were rehydrated with "`datasets rehydrate --directory ncbi_dataset`". The

quality of samples was measured by inspecting NCBI-reported CheckM completeness, contamination, and count of consensus-assembled genomic regions (contigs) (Supplementary Fig.4, 8, 13). The failure to download and analyze 4,660 samples (Supplementary Fig.1) was primarily due to obsolete genomes in the NCBI database, taxonomic assignment errors that caused annotation failures, or other technical issues.

Please note that the choice of reference database and taxonomy impacts the annotation results. We decided to adhere to the taxonomy classification provided by GTDB and their more stringent taxonomy verification schema, which we treat as a gold standard for this work. For additional dataset details, please refer to the GTDB release 207 website. However, biological databases are continuously updated as new bacterial species are discovered, particularly in the context of metagenome-assembled genomes (MAGs) and environmental samples. For example, within the year 2021, 12,556 new candidate species-level operational taxonomic units (OTUs) were identified[25]. The GTDB database also experienced a significant increase, adding 17,809 Bacteria and Archaea species between release-202 in 2021 and release-207 in 2022 (Supplementary Fig.21). Additionally, the taxonomic assignments for each genome are registered between GTDB releases, which allows for a history of assignments across newer and older releases.

Escherichia coli

E. coli samples were derived by extracting part of the string that starts with “s__” from the database key “`gtdb_taxonomy`” with command “`str_extract(bac120_r207$gtdb_taxonomy”, "s__[a-zA-Z\\s]+"`) from string R library. The output was queried for “`s__Escherichia coli`”.

The analysis included 24,393 *E. coli* samples, with an average completeness of 99.72% (SD = 1.11%) and contamination levels averaging 0.41% (SD = 0.69%). The mean number of contigs per assembly was 185.04 (SD = 149.54), with a genome size averaging 5,120,392 base pairs (SD = 274,048 bp). Variations in genome size among *E. coli* samples likely result from assembly issues, samples derived from

metagenomes, or natural intraspecies variability[26]. From the 24,393 genomes analyzed, we identified 2,073 distinct serotypes based on their O- and H-types in 20,525 genomes, where in 10 samples serotyping failed. Among the remaining genomes, 3,336 had only an H-type, 358 had only an O-type, and 164 had neither (Supplementary raw data).

Archaea and Bacteria

The representative group for both Bacteria and Archaea files was queried with database key "`gtdb_representatitive`" for value "t" for GTDB representative genomes.

The dataset comprised 30,107 Bacteria and 2,791 Archaea samples (Supplementary Fig.2-3). Bacteria showed an average completeness of 95.45% (SD = 9.12%), while Archaea had a completeness of 85.57% (SD = 13.02%). Contamination levels averaged 1.03% (SD = 1.34%) for Bacteria and 1.28% (SD = 1.58%) for Archaea. The mean number of contigs per assembly was 93.38 (SD = 154.05) for Bacteria and 113.9 (SD = 128.18) for Archaea. Genome sizes averaged 4,009,159 bp (SD = 2,120,640 bp) for Bacteria and 1,895,897 bp (SD = 1,042,641 bp) for Archaea. Detailed dataset information is available in Supplementary Fig.7-8 (Bacteria), Fig.12-13 (Archaea) and Supplementary raw data.

Overall, our preliminary dataset statistics indicate that the genome assemblies were sourced from high-quality samples, with minimal potential contamination, making them suitable for downstream analysis.

Frameshifts

To simulate genomes of lower assembly quality and the effects of open reading frame shifts on annotation quality, we modified the obtained FASTA files of the samples by inserting pseudo-random deletions. In the analysis, we use 0.5%, 1%, and 2% deletions of nucleotides in a sample's FASTA file. The frameshift analysis was performed on a 32,914-genome subset of archaea, bacterial and Escherichia coli samples, generating in total 98,742 data points. Considering the assumption of equal spread of deletion events and an average bacterial genome length of

4,372,147 bp in our dataset, we anticipate deletion occurrences at regular intervals of 199, 99, and 49 bp, respectively.

The expected fragment length between deletions was calculated via

$$Deleted\ Nucleotides\ = \frac{Percentage\ Deleted}{100} * \ Original\ Genome\ Length$$

$$Number\ of\ Fragments\ = \frac{Original\ Genome\ Length - Deleted\ Nucleotides}{Deleted\ Nucleotides} + 1$$

$$Average\ Fragment\ Length\ = \frac{Total\ Remaining\ Genome\ Length}{Number\ of\ Fragments}$$

The random deletions are generated with the following code stored in https://github.com/AggresiveHayBale/anno_bench in script “noise.sh”.

Metagenome-assembled Genomes

We recognize a genome as a metagenome if the key “*ncbi_genome_category”* has a value “*derived from metagenome*". The dataset comprises 101 samples designated as metagenome-assembled genomes in the *E. coli* group, 3,137 in the bacterial group, and 2,031 in Archaea. The *E. coli* and Archaea metagenomic samples were excluded from metagenome analysis due to low counts (*E. coli*) and significant differences in tool performance compared to the other groups (Archaea).

Additional parameters

The NCBI submission date was extracted from the key “*ncbi_date*” located in the dataset file. The used dataset consists of samples submitted between 2000 and 2021 (Supplementary Fig.23). The NCBI-reported protein count is derived from the key “*ncbi_protein_count*” while PGAP annotation is the total count of CDS from *de novo* annotation done for this benchmark.

Annotation and tool description

We recognize that specialized single-function tools, that depend on previously detected feature starts and ends, like DEEP-FRI[27], may outperform generalist tools in specific tasks.These tools were excluded from this analysis due to their specialized focus on either a singular step of describing previously detected features

or the annotation of specific subsets of features. Furthermore, popular tools like RAST[28], which rely on external servers, were excluded from our benchmark due to technical limitations imposed by the use of external, not self-managed servers, thereby limiting the potential for a large-scale benchmark. The selected tools use a varied approach to CDS feature prediction using Prodigal (Prokka, EggNOG-mapper), Prodigal with improved short ORF detection (Bakta), and ORFfinder with a large set of other tools (PGAP). For a detailed description of each tool, please refer to the original articles.

Tools

Tools were used with the suggested default parameters, as outlined in the respective GitHub help pages. In the case of EggNOG-mapper, the parameters of the default setting for the online version of the tool (http://eggnog-mapper.embl.de) were used for the “Genome” data type and Gene prediction method “Prodigal” and “Diamond” search method.

Genetic code translation table

For *Mycoplasmatota* (formerly *Tenericutes*), the 4th translation table was used instead of the standard one[29]. Using flags:

- Prokka: `–gcode 4`
- Bakta: `–translation-table 4`
- EggNOG-mapper: `--trans_table 4`
- PGAP does not require a flag as it chooses the translational table using the species name to make a decision

Prokka

Containerized Prokka 1.14.6 (DockerHub container: nanozoo/prokka:1.14.6--c99ff65) was executed with command:

```
prokka --compliant --outdir output --force --prefix
${name}_prokka --quiet ${fasta}
```

The Prokka database used for the annotation is specific to the Prokka version and is an integral part of its GitHub repository. Prokka and its bundled database were the oldest of all the investigated tools, with the last update in November 2019 (v1.14.5; https://github.com/tseemann/prokka/releases). This indicates that the functional annotations provided by Prokka may not be as current as those offered by other tools. Prokka is a popular, fast, and lightweight tool suitable for both small-scale system requirements and low storage needs (database size: 0.6 GB), developed in 2014 by Torsten Seemann for draft genome annotations. Prokka employs as its core databases ISfinder, NCBI Bacterial Antimicrobial Resistance Reference Gene Database, and UniProtKB (SwissProt), followed by bacteria-specific HMM libraries such as Pfam and TIGRFAMs for annotation of predicted CDS features[12].

Bakta

Containerized Bakta 1.7 (Docker Hub container: nanozoo/bakta:1.7.0--23ae510) was executed with command:

```
bakta --prefix ${name}_bakta --db database/
--keep-contig-headers ${fasta}
```

The annotation was executed with the database version 5.0.

Bakta, while taking inspiration from Prokka, offers more advanced taxon-independent annotation capabilities while retaining Prokka's user-friendly simplicity. Bakta had 40 releases between November 2020 and July 2024 (v1.9.4; https://github.com/oschwengers/bakta/releases), while its database had 8 releases (https://zenodo.org/records/10522951), with the most recent on January 19, 2024 (v5.1). Bakta requires a larger database than Prokka (60 GB) and has higher hardware requirements. It also provides a web frontend for small-scale annotation. Bakta annotates features such as tRNA, tmRNA, rRNA, ncRNA genes, CRISPR, CDS, pseudogenes, ncRNA cis-regulatory regions, oriC/oriV/oriT, and assembly gaps. The database used for annotation incorporates UniProt's UniRef100 clusters, UniParc, and NCBI RefSeq sequences, with UniProt's UniRef90 seed sequences as a fallback. Bakta also integrates external databases such as NCBI AMRFinderPlus, COG, DoriC, ISFinder, Mob-suite, Pfam, Rfam, RefSeq, UniProtKB/Swiss-Prot, and VFDB. Functional annotation in Bakta includes KEGG pathways, PFAM protein

families, RFAM RNA families, enzyme commission numbers, Clusters of Orthologous Groups, and Sequence Ontology terms.

EggNOG-mapper

Containerized EggNOG-mapper 2.1.9 (Docker Hub container: nanozoo/eggnog-mapper:2.1.9--4f2b6c0) was executed with command:

```
emapper.py -i ${fasta} -m diamond --data_dir ${eggnog_db_dir}
--itype genome --genepred prodigal --dmnd_ignore_warnings
--dbmem --translate --go_evidence non-electronic
--pfam_realign none --report_orthologs --decorate_gff yes
--evalue 0.001 --score 60 --pident 40 --query_cover 20
--subject_cover 20 --tax_scope auto --target_orthologs all -o
${name}_eggnog
```

We used the respective version of the database from November 22, 2022.

EggNOG-mapper is unique in its focus on orthology, allowing for the annotation of both prokaryotes and eukaryotes, unlike other tools that target only prokaryotes and employ a homology approach. Similar to Bakta, it provides a web frontend for small-scale annotation. EggNOG-mapper had 19 releases from November 2017 to August 2023 (v2.1.12; https://github.com/eggnogdb/eggnog-mapper/releases). EggNOG-mapper requires a similar large database to Bakta (51 GB). Its database had 11 releases, but the latest (v6.0.0) is not implemented in the tool available on GitHub, which uses version 5.0.2 from March 2021. It exclusively annotates gene type features, excluding any other type (rRNA, tRNA). EggNOG-mapper annotations are derived from their internal EggNOG database[30]. The functional annotation by EggNOG-mapper includes categories such as CAZy, COG, COG Categories, Enzyme Commission numbers, KEGG orthologs, BRITE hierarchies, BiGG Reactions, KEGG reaction classes, KEGG transporter classifications, KEGG pathways, KEGG reactions, KEGG modules, and PFAMs.

PGAP

Containerized PGAP 2022-10-03.build 6384 (Docker Hub container: ncbi/pgap:2022-10-03.build6384) was executed with command:

```
cwltool /pgap/pgap/pgap.cwl --fasta ${fasta}
--ignore_all_errors --report_usage --submol meta.yaml
```

With the respective version of the database accessed on February 7, 2023. The taxonomic information (species name) in the “meta.yaml” file is derived from the GTDB database “ncbi_taxonomy” key.

PGAP has been in constant development by NCBI since 2001 and is internally used for annotating NCBI databases. PGAP had 31 releases from July 2018 to July 2024. Before the GitHub releases, updates were announced on the NCBI website, with 20 versions released between 2013 and 2018 (https://www.ncbi.nlm.nih.gov/refseq/annotation_prok/release_notes/). PGAP has a smaller database compared to Bakta and EggNOG-mapper, however larger than Prokka (30.4 GB). Since PGAP updates its database with each release, it is challenging to determine the exact number of database updates that have occurred. However, manual inspection revealed that 14 GitHub releases mentioned database changes, with an average interval of 123 days, making PGAP the tool with the most frequent database updates. The main distinguishing feature is the requirement of the tool for taxonomic information, tailoring the annotations to genome-specific patterns, and reducing the risk of human errors (wrong translational table use, detection of contamination, or wrong species assignment). The primary drawback of the tool is its relatively high computational requirements and the need for metadata preparation. PGAP distinguishes the following types of features: Gene, CDS, rRNA, tRNA, ncRNA, and CRISPR. It utilizes multiple data sources like TIGRFAM, Pfam, Rfam, PRK, NCBIfams, BlastRules, and several in-house created databases.

Workflow execution

The tools were run in a managed computing service environment on the Google Cloud Platform infrastructure, using the Google Cloud Batch executor. This was accomplished through a custom-built Nextflow[31] pipeline (https://github.com/AggresiveHayBale/anno_bench) and Docker containers (Supplementary Fig.23). The virtual hardware specifications used for tool execution are as follows:

1. Prokka

    Docker Hub container: nanozoo/prokka:1.14.6--c99ff65

    CPUs: 4

    Memory: 2 GB

2. Bakta

    Docker Hub container: nanozoo/bakta:1.7.0--23ae510

    CPUs: 8

    Memory: 14 GB

3. EggNOG-mapper

    Docker Hub container:: nanozoo/eggnog-mapper:2.1.9--4f2b6c0

    CPUs: 6

    Memory: 48 GB

4. PGAP

    Docker Hub container: ncbi/pgap:2022-10-03.build6384

    CPUs: 8

    Memory: 8 GB

The median RAM utilization (Physical memory usage) and median CPU time (Task execution real-time) were read from the Nextflow execution report.

Data analysis

In contrast to similar studies[11], we focus on the measurement of coding space. Coding space denotes the sum of lengths of all detected features. Features length is defined by subtracting the stop position from the start position in the annotation file. The described coding space denotes the coding space of all features with feature description, recognized by a string starting with “product=”(Prokka, PGAP), ”Name=” (Bakta), “em_desc=” (EggNOG-mapper), and ending with “;” in all of the tools. The proportion of the above-mentioned coding spaces is a proportion of the relative

category relative to the total genome size reported by the NCBI in the metadata field "ncbi_genome_size".

$$Gene\ Length\ (L) = Stop\ Position - Start\ Position$$

$$Total\ Coding\ Space = \sum_{i} L_i$$

$$Described\ Coding\ Space = \sum_{i-j} L_{i-j}$$

$$Undescribed\ Coding\ Space = \sum_{j} L_j$$

$$Proportion\ of\ Coding\ Space = \frac{\sum_{i} L_i}{NCBI\ Genome\ Size}$$

Where i denotes the feature count and j denotes the undescribed feature count.

We prefer coding space metric over the total feature count to minimize the risk of artificially inflated or reduced gene counts stemming from differences in how tools merge and split subunits of genes or report only fragments of a complete feature. This approach helps mitigate such biases. However, due to the limited occupied coding space and the conservative nature of RNA subgroups, we use the raw count for comparing these type of features.

When the undescribed feature proportion is used, the proportion of hypothetical proteins is calculated as follows:

$$Proportion\ of\ Undescribed\ Proteins\ (\%) = \frac{\sum Undescribed\ feature\ count}{N} * 100\%$$

Where N denotes the overall feature count.

GO coverage and richness were calculated in the following manner:

$$GO\ coverage\ (\%) = \frac{\sum Features\ with\ at\ least\ one\ GO\ term}{N} * 100\%$$

$$GO\ richness = \frac{\sum GO\ terms\ per\ gene}{N}$$

Where N denotes the overall feature count.

The original and modified sequences (frameshifts) are then annotated with Prokka, Bakta, EggNOG-mapper, and PGAP. The created annotation files in .gff3 (Bakta) and .gff (Prokka, EggNOG-mapper, PGAP) format are passed to the R language script that performs an automated count of the features and preliminary analysis.

Annotations were analyzed with R[32] (4.2.1) with Tidyverse[33] packages (tidyr (1.2.1), dplyr (1.0.10), readr (2.1.3), and stringr (1.4.1)). Additional data visualization was performed using following R libraries: viridis(0.6.3), waffle(0.7.0), hrbrthemes(0.8.6), RColorBrewer (1.1-3), ggpubr(0.6.0), gridExtra (2.3), ggplot2[33] (3.4.2), nvimcom (0.9-147), and geneviewer (0.1.11).

The features were treated as undescribed if Prokka, Bakta, and PGAP described them as “hypothetical protein”, or in the case of EggNOG-mapper, if the product does not have a description (denoted as “None” or “NA”).

RNA features are recognized by extracting the “rRNA”, “tmRNA”, and “tRNA” from the 3rd column of the main annotation file (.gff,.gff3) in all of the tools.

GO terms and other functional annotation terms were extracted by querying the 9th column of the main annotation file, in case of

Prokka by: “GO”, “COG[0-9]{1,4}”

Bakta: "GO:", "KEGG:", "PFAM:", "RFAM:", "EC:", "COG:COG", "COG:(?!COG)" (COG group), "SO:"

EggNOG-mapper The extracted terms are: "em_CAZy", "em_COG_cat", "em_EC", "em_KEGG_ko", "em_BRITE", "em_BiGG_Reaction", "em_KEGG_rclass", “em_KEGG_TC", "em_KEGG_Pathway", "em_KEGG_Reaction", “em_KEGG_Module", "em_PFAMs".

PGAP: “GO:”

The proportion of proteins with at least one GO term was calculated via

$$Term\ proportion\ = \frac{F_{x \geq 0}}{i} * \ 100$$

Where *F* denotes features, *x* is the count of terms per feature, and *i* is the total feature count.

The serotyping was done using ECTyper[34] (1.0.0) using the command: `ecoli_serotyper.py -i ${fasta}.`

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